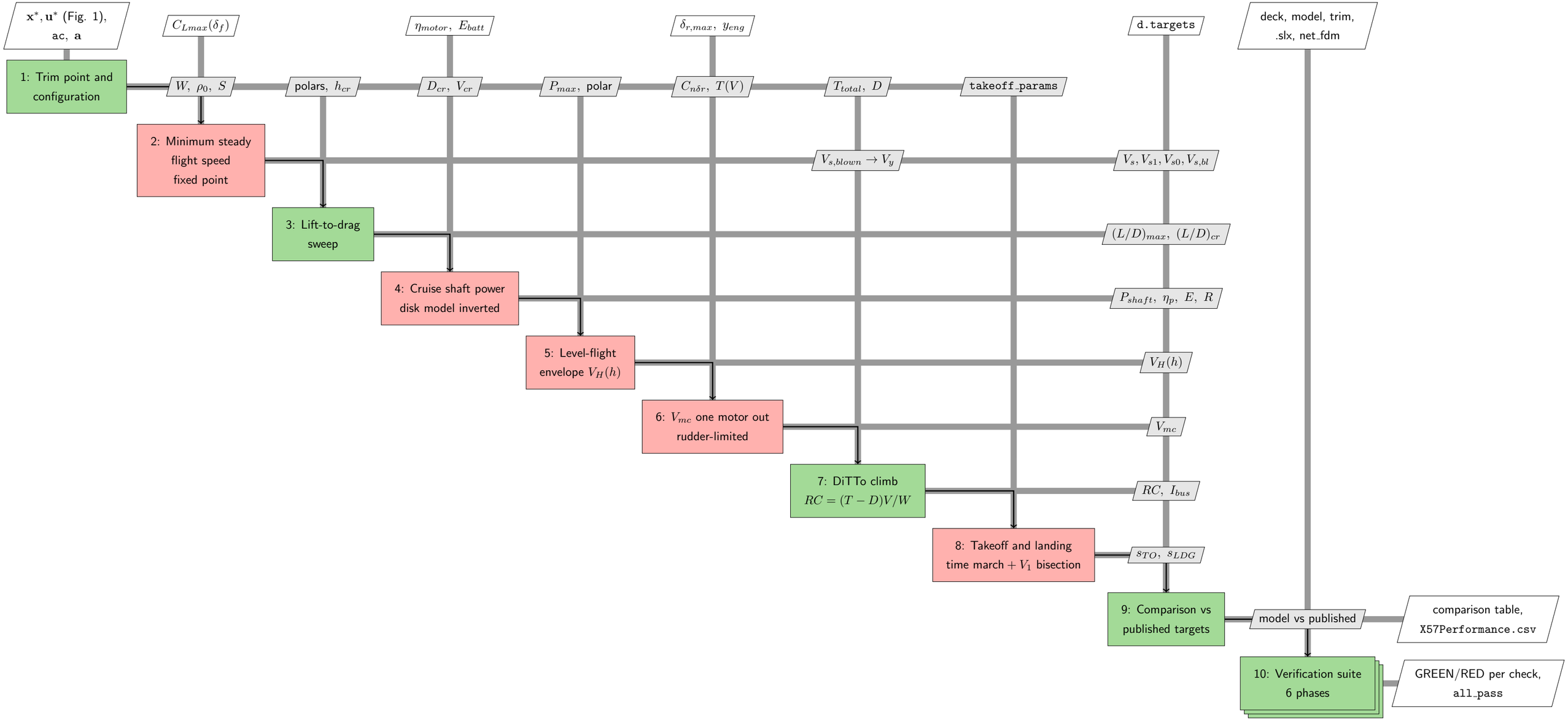

$\mathbf{x}^*, \mathbf{u}^*$ (Fig. 1), ac, $\mathbf{a}$
$C_{Lmax}(\delta_f)$
$\eta_{motor},\ E_{batt}$
$\delta_{r,max},\ y_{eng}$
d.targets
deck, model, trim, .slx, net_fdm
1: Trim point and configuration
$W,\ \rho_0,\ S$
polars, h_{cr}
$D_{cr},\ V_{cr}$
P_{max}, polar
$C_{n\delta r},\ T(V)$
$T_{total},\ D$
takeoff_params
2: Minimum steady flight speed fixed point
$V_{s,blown} \to V_y$
$V_s, V_{s1}, V_{s0}, V_{s,bl}$
3: Lift-to-drag sweep
$(L/D)_{max},\ (L/D)_{cr}$
4: Cruise shaft power disk model inverted
$P_{shaft},\ \eta_p,\ E,\ R$
5: Level-flight envelope $V_H(h)$
$V_H(h)$
6: V_{mc} one motor out rudder-limited
V_{mc}
7: DiTTo climb $RC = (T - D)V/W$
$RC,\ I_{bus}$
8: Takeoff and landing time march + V_1 bisection
$s_{TO},\ s_{LDG}$
9: Comparison vs published targets
model vs published
comparison table, X57Performance.csv
10: Verification suite 6 phases
GREEN/RED per check, all_pass